Final paper

Alyssa Hagele, Abby Malech, Henderson Vo

Introduction

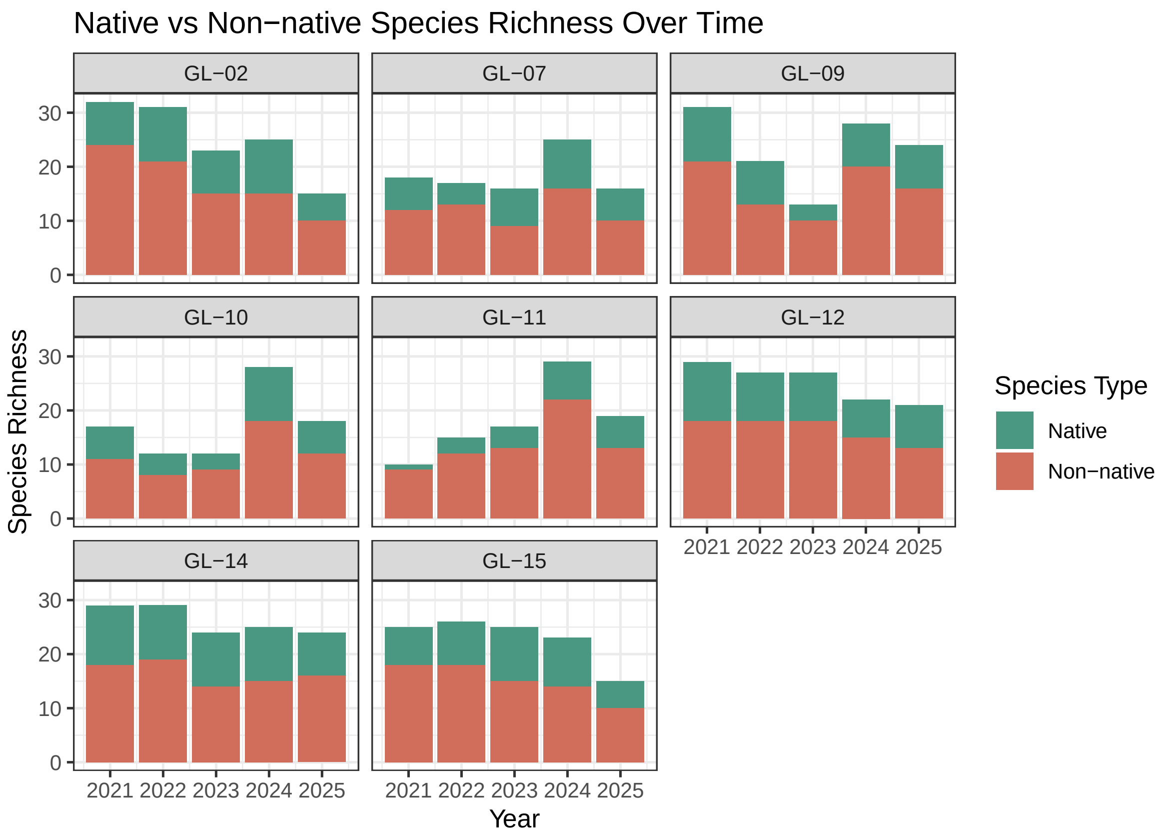
Methods

To answer our research question, we used a vegetation dataset sourced from the Cheadle Center for Biodiversity and Ecological Restoration (CCBER), and provided to us by the ENVS 193DD course. This dataset spans 5 years (2021-2025) of vegetation data from across several habitats in North Campus Open Space (NCOS) including grasslands, mosaic habitats, wetlands, open water, woodlands, and shrublands.

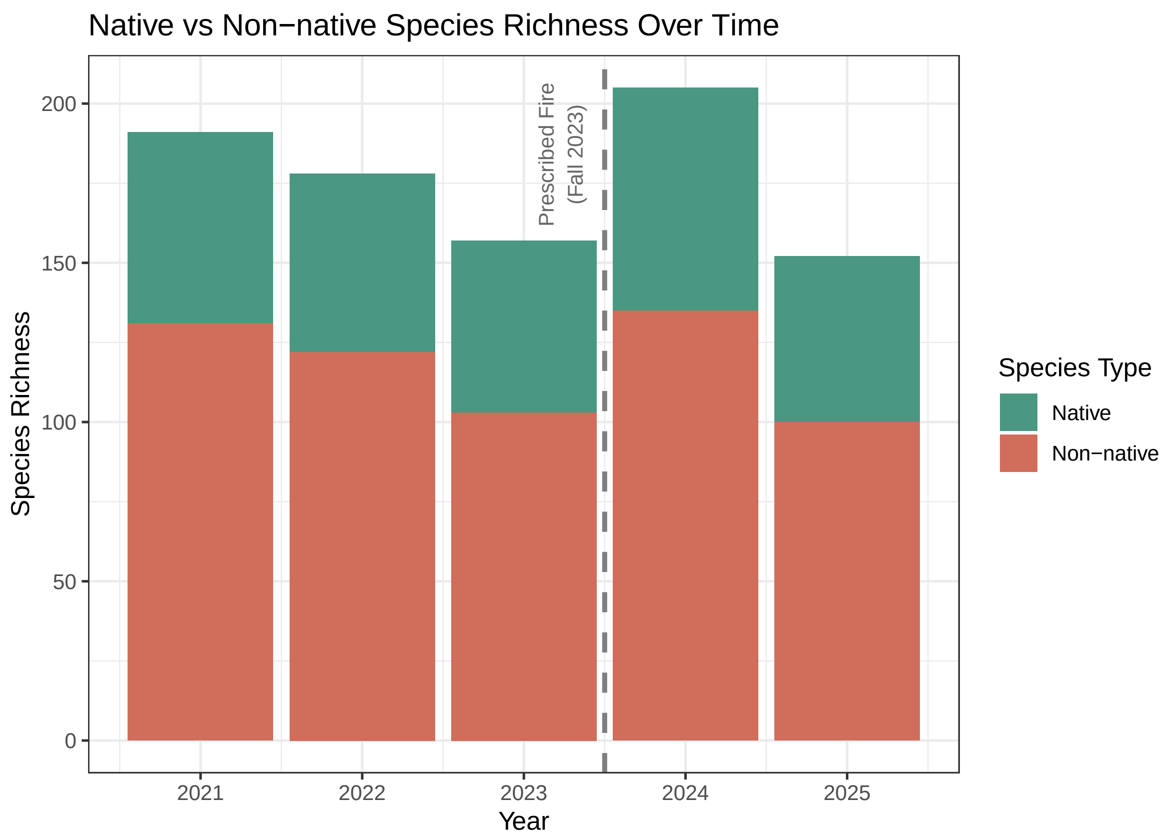
We chose to focus on native perennial grasslands, which encompasses measurements from 8 different transects across the 16.8-acre mesa. The method of data collection used for vegetation surveys is the Quadrat Transects (QT) method. This method involves the use of 11 one-square-meter quadrats placed by a 30-meter transect, with quadrats alternating between the left and right side of the transect line every 3 meters. "The quadrats are subdivided into 100 ten-centimeter squares and the percent cover is estimated [by human monitors] for each species in the quadrat," (Rickard 2024).

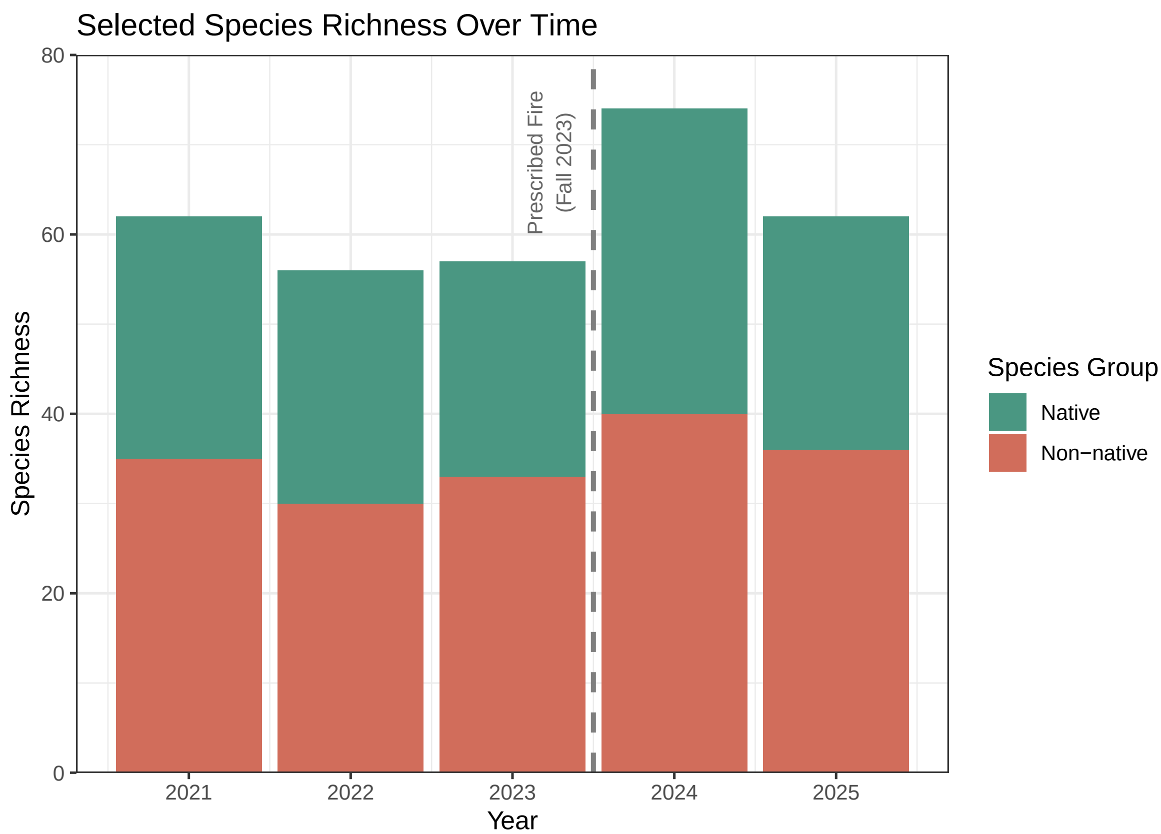
We will not be performing a formal statistical analysis for our project because we are primarily exploring a correlative relationship rather than predicting trends for species cover and richness. However, we will be identifying major events that might influence a directional response in vegetation percent cover and species richness, and examining whether a response was captured in our dataset. Major events include the prescribed burn in Fall 2023 because of its effects as an intermediate disturbance event, as well as wet (2023, 2024) and dry (2021, 2022) water years because water is a limiting resource for key native grass species (S. pulchra). These factors are potential influences on how percent cover and species richness change over time.

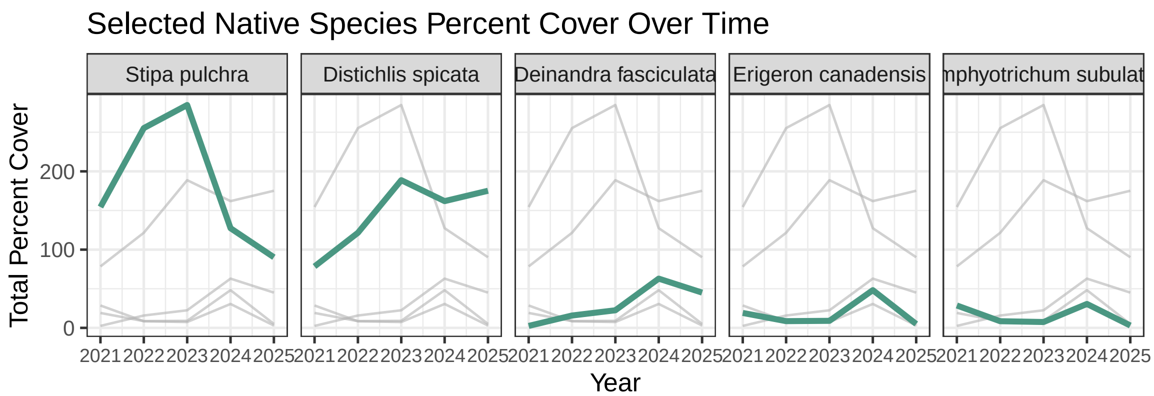
Results

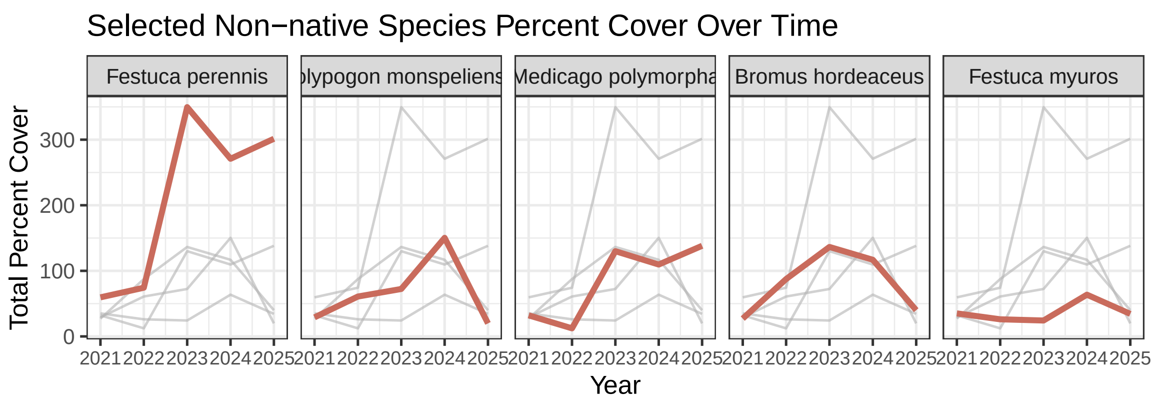


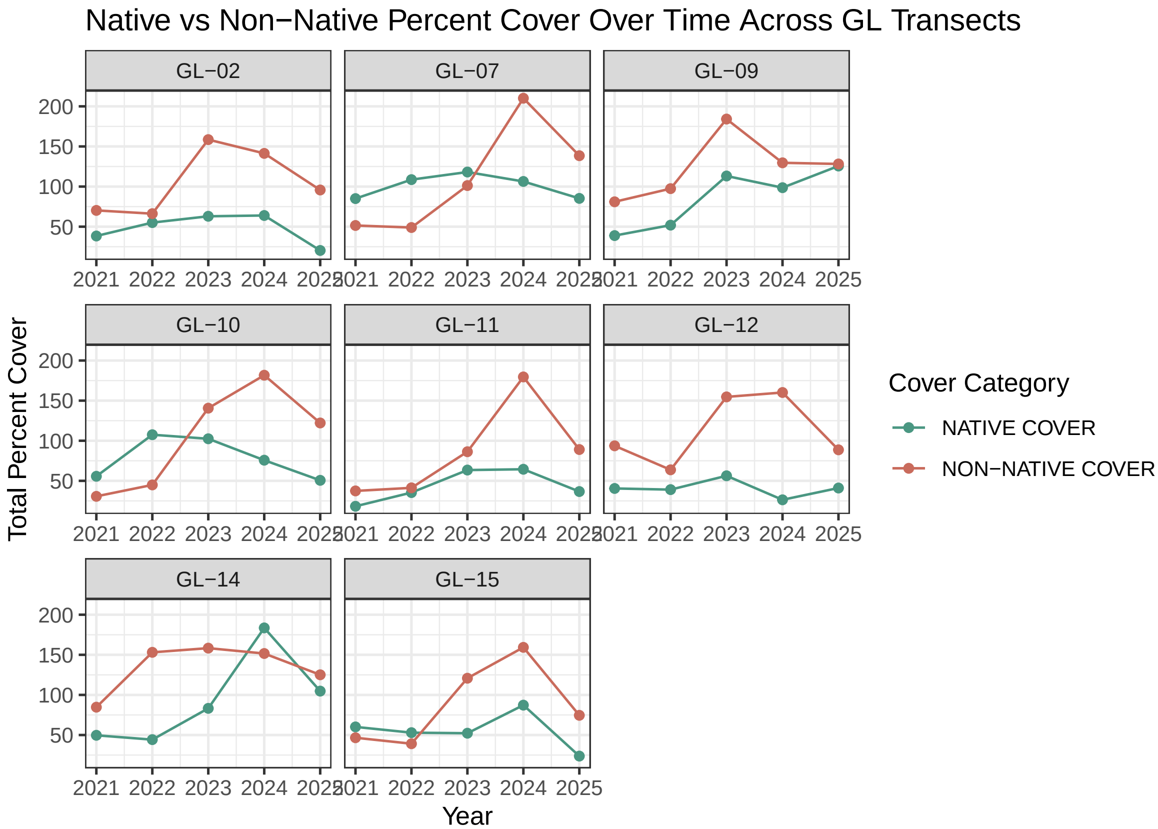
Native vs Non−native Species Richness Over Time
GL−02
GL−07
GL−09
GL−10
GL−11
GL−12
GL−14
GL−15
0
10
20
30
2021 2022 2023 2024 2025
Species Richness
Year
Species Type
Native
Non−native

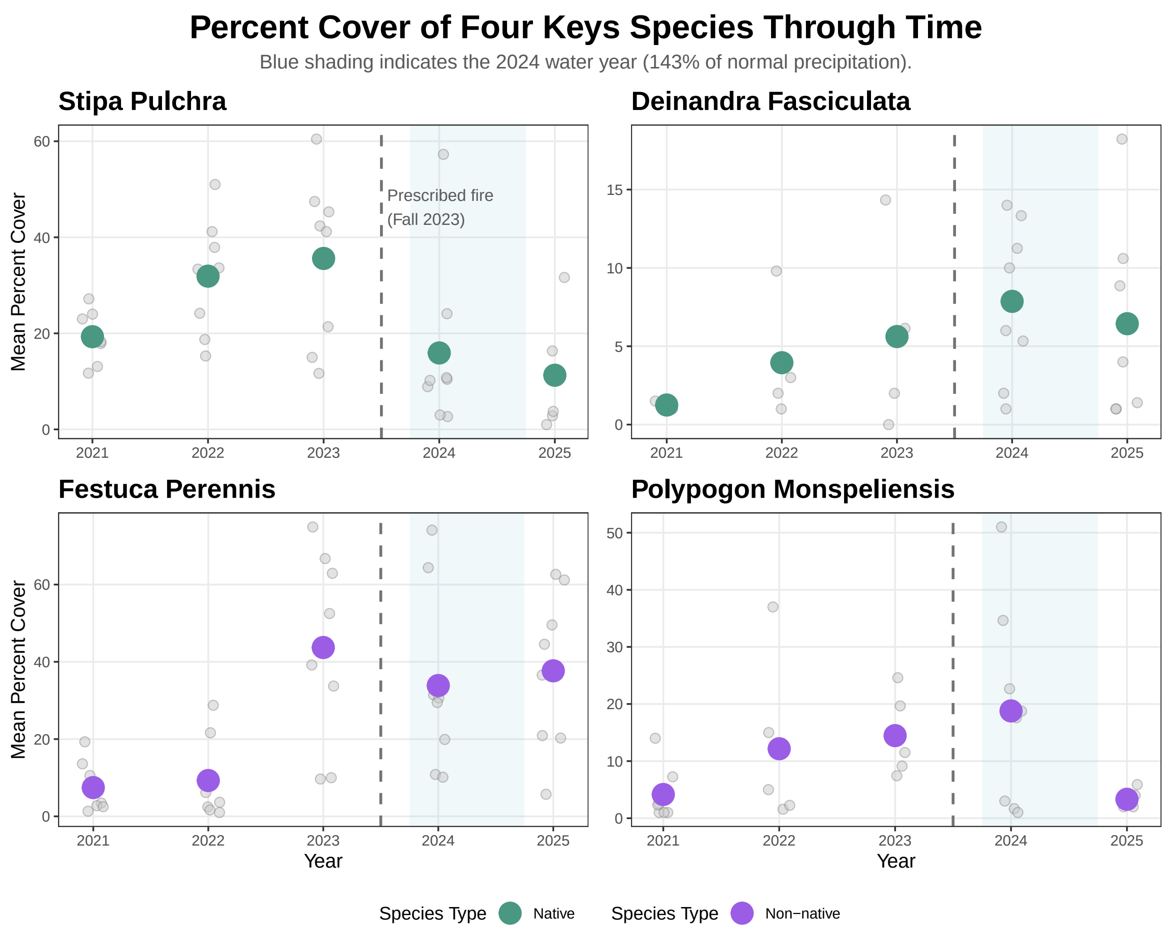

Native vs Non−native Species Richness Over Time
Prescribed Fire
(Fall 2023)
Species Richness
0
50
100
150
200
2021
2022
2023
2024
2025
Year
Species Type
Native
Non−native


Selected Species Richness Over Time
Species Richness
80
60
40
20
0
Prescribed Fire
(Fall 2023)
2021
2022
2023
2024
2025
Year
Species Group
Native
Non−native


Selected Native Species Percent Cover Over Time
Stipa pulchra
Distichlis spicata
Deinandra fasciculata
Erigeron canadensis
mphyotrichum subulat
Total Percent Cover
0
100
200
2021
2022
2023
2024
2025
Year


Selected Non−native Species Percent Cover Over Time
Festuca perennis
olypogon monspeliens
Medicago polymorpha
Bromus hordeaceus
Festuca myuros
Total Percent Cover
0
100
200
300
2021
2022
2023
2024
2025
Year


Native vs Non−Native Percent Cover Over Time Across GL Transects
GL−02
GL−07
GL−09
GL−10
GL−11
GL−12
GL−14
GL−15
Total Percent Cover
Year
200
150
100
50
2021
2022
2023
2024
2025
Cover Category
NATIVE COVER
NON−NATIVE COVER



Discussion

References

Rickard, A. 2024. "North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024." *UC Santa Barbara: Cheadle Center for Biodiversity and Ecological Restoration*. https://escholarship.org/uc/item/16p046mb.